Practical 8:

Aim : A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Code:

```
~
  GNU nano 8.7
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frame[4], time[4];
    int i,j,n=12,frames=4,hit=0,fault=0,counter=0,pos,min;

    for(i=0;i<frames;i++)
        frame[i] = -1;

    for(i=0;i<n;i++) {
        int found = 0;

        for(j=0;j<frames;j++) {
            if(frame[j] == pages[i]) {
                hit++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }

        if(!found) {
            fault++;

            if(i < frames) {
                frame[i] = pages[i];
                counter++;
                time[i] = counter;
            }
            else {
                min = time[0];
                pos = 0;

                for(j=1;j<frames;j++) {
                    if(time[j] < min) {
                        min = time[j];
                        pos = j;
                    }
^G Help          ^O Write Out     ^F Where Is      ^K Cut           ^T E
^X Exit          ^R Read File     ^\ Replace       ^U Paste         ^J J
```

```
                    }
                }

                frame[pos] = pages[i];
                counter++;
                time[pos] = counter;
            }
        }
    }

    printf("Hits = %d\nFaults = %d\n", hit, fault);
    printf("Hit %% = %.2f\nFault %% = %.2f\n",(float)hit/n*100,(float)fault/n*100)

    return 0;
}

                                                    [ Soft wrapping of overlon
^G Help        ^O Write Out    ^F Where Is     ^K Cut          ^T Execute
```

Output:

```
~

hp@DESKTOP-CA6S8T8 MSYS ~
$ nano lru.c

hp@DESKTOP-CA6S8T8 MSYS ~
$ gcc lru.c -o lru

hp@DESKTOP-CA6S8T8 MSYS ~
$ ./lru
Hits = 4
Faults = 8
Hit % = 33.33
Fault % = 66.67

hp@DESKTOP-CA6S8T8 MSYS ~
$ |
```